

CSE340: Computer Architecture

Assignment 2

Chapter 2

Question - 1:

Given a machine code, what do we understand by looking at the **Opcode** field?

Question - 2:

Suppose you have converted LD X9, 10[X21] to its equivalent RISC-V machine code. Now based on which **field** machine will understand the size of the data being loaded into register X9?

Question - 3:

```
if ( A[3] != A[6]){
    if (A[3] == 0) {
        A[3] = A[3] + g;
    }else{
        A[6] = A[6] / 16;
    }
}
else{
    A[6] = B[6] * 9
}
```

a) Construct the equivalent RISC-V code of the above mentioned C code.

b) Base addresses of array A and B are in register X20 and X21. The value of variable g is stored in register X18.

From your converted codes, pick any instruction from each of the three formats (R, I, S) and convert them into their corresponding binary values.

Question - 4:

$A[B[i]] = g + B[5]$

Construct the equivalent RISC-V code of the above mentioned C code. Base addresses of array A and B are in register X20 and X21. The value of variable g is stored in register X18.

Question - 5:

What is the significance of the funct3 field in case of a Load instruction?